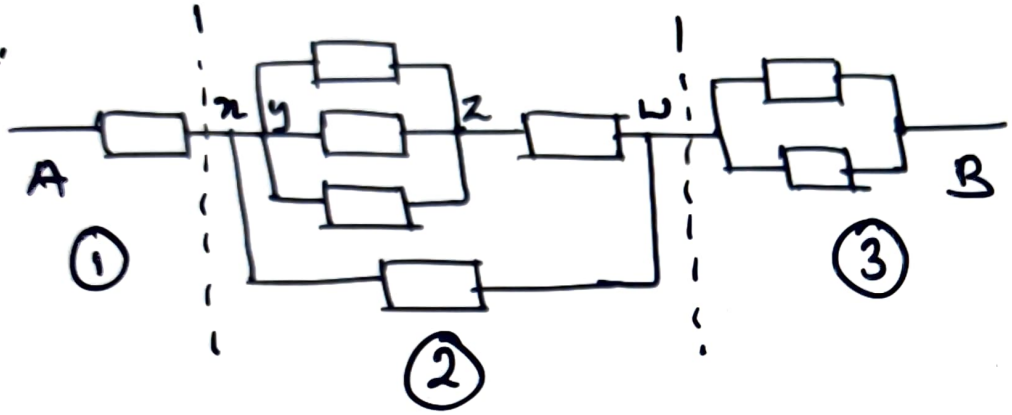
Tutorial 1 : MIT 6.041 Probability Systems Analysis

1. $A \subset B \Rightarrow A \cap B = A \Rightarrow P(A \cap B) = P(A) = P(A)P(B)$
only if $P(A) = 0$ or $P(B) = 1 \Rightarrow A = \phi$ or $B = \Omega$

2.



$P(\text{subsystem 1 is operational}) = P(A \to x) = p$

$P(y \to z) = 1 - (1-p)^3$; $P(z \to w) = p$

$P(x \to w) = 1 - (1-p)(1 - p(1-(1-p)^3))$

$P(\text{subsystem 2 is operational}) = 1 - (1-p)(1 - (1-(1-p)^3)p)$

$P(\text{subsystem 3 is operational}) = P(w \to B) = 1 - (1-p)^2$

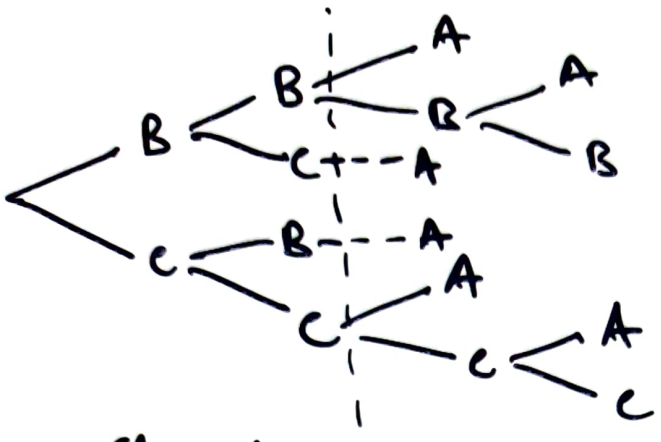
$\Rightarrow P(\text{system is operational}) = P(A \to x)\,P(x \to w)\,P(w \to B)$

$= p(1-(1-p)(1-(1-(1-p)^3)p))(1-(1-p)^2)$

3. Let A/B/C : Al/Bo/Ci win a game

All games are independent

(a) i) 2nd round will be required if outcomes in stage 1 are either BB or CC



Stage 1 Stage 2

$\Rightarrow P(\text{2nd round}) = P(BB) + P(CC) = 0.6^2 + 0.4^2 = 0.52$

ii) Bo wins 1st round

$= P(BB) = 0.36$

iii) P(A retains his championship)

$= P(\{BC, CB\}) + P(\{BBA, BBBA, CCA, CCCA\})$

$= 2 \times 0.6 \times 0.4 + 0.6^2 \times 0.5 + 0.6^2 \times 0.5^2 + 0.4^2 \times 0.7$

$+ 0.4^2 \times 0.3 \times 0.7$

$= 0.48 + 0.36 \times 0.75 + 0.16 \times 0.91 = 0.8956$

(b) i) Bo is surviving challenger : $\{BBA, BBBA, BBBB\}$

$P(\text{Bo is surviving challenger} \mid \text{2nd round})$

$= \dfrac{P(\text{Bo is surviving challenger} \cap \text{2nd round})}{P(\text{2nd round})}$

$= \dfrac{P(\{BBA, BBBA, BBBB\})}{P(\{BB, CC\})} = \dfrac{0.36}{0.52} = 0.6923$

ii) A retains championship $\cap$ 2nd round $= \{BBA, BBBA, CCA, CCCA\}$

P(A retains championship | 2nd round)

$$= \frac{P(\{BBA, BBBA, CCA, CCCA\})}{P(\{BB, CC\})} = \frac{0.4156}{0.52}$$

$$= 0.7992$$

(c) 2nd round was required and had only 1 game

$$= \{BBA, CCA\}$$

2nd round was required and had only 1 game ∩ B0 won 1st round $= \{BBA\}$

$$\Rightarrow \text{Probability} = \frac{P(\{BBA\})}{P(\{BBA, CCA\})} = \frac{0.6^2 \times 0.5}{0.6^2 \times 0.5 + 0.4^2 \times 0.7}$$

$$= \frac{0.18}{0.292} = 0.6164$$